Project 7: Complex systems and classical systems biology

MCSB Bootcamp — Mathematical and Computational Track

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Consider the simplified Guido–Collins system with an activator transcription factor and an inhibitor transcription factor.

You may use the code in the Synthetic biology, simplified notebook, section 2.3. Its `dxdt` already takes both `C`, the activator concentration, and `I`, the inhibitor concentration.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp
```

```
# The model, copied from the Synthetic biology, simplified notebook, section 2.3.
```

Part 1: The effect of the inhibitor

Assume the concentration of inhibitor is fixed at 10 uM.

Plot the steady-state amount of protein as a function of the activator concentration, but make the axis showing concentration of activator a logarithmic axis.

What is the concentration of activator that results in half its maximal effect? This is called the EC50 of the activator.

```
# Your code here.
```

Part 2: The effect of the activator

Assume the concentration of activator is fixed at 10 uM.

Plot the steady-state amount of protein as a function of the inhibitor concentration.

What is the concentration of inhibitor that results in half its maximal inhibition? This is called the IC50 of the inhibitor.

```
# Your code here.
```